



Exercise 1

You are tasked with purchasing one or more new computers for private or company use, workstations at the University Computer labs, Name criteria you can use to evaluate computer performance.

Exercise 2

Explain the following terms: CPU, ALU, RAM, ROM, BIOS, MIPS, FLOPS?

Exercise 3

Calculate the maximum possible data transfer rate in MiB/s for a 64-bit wide bus running at 80MHz.

Exercise 4

You have a 32-bit wide bus running at 40MHz. The bus protocol takes up 10% of the bandwidth (Overhead) to assure synchronization and data integrity. What is the net data rate in MiB per second?

Exercise 5

A computer has 32-bit wide registers, 32 data lanes and 21 address lanes.

- What is the size of a register in bytes?
- What is the largest possible binary integer number that can be stored in a register?
- How large is the address space, i.e., how many memory cells can be addressed?
- What is the maximum RAM in MiB assuming the memory is addressed byte-wise?
What if we address it word-wise (i.e., each memory cell contains 32 bits)?

Exercise 6

Assume that the CPU of a hypothetical computer has

- a data bus of width 8 bits,
- an address bus of width 12 bits,
- a control bus of width 6 bits,
- working registers of width 32 bits.

The buses and CPU are all clocked the same at 1 MHz.

- What is the maximum possible amount of RAM in KiB, assuming that the memory is addressed byte-wise?
- How many microseconds will it take to populate a working register with a 32-bit value from RAM, assuming that no bandwidth is required by the bus protocol to synchronize and secure the data transmission (no overhead)?